

Greece

Hellenic Competition Commission

The Hellenic Competition Commission ('HCC') in the past two years has made a leap forward regarding its technological-digital structure while it is becoming a leader, among the EU Competition Authorities, in the use of computational means in its investigations, already designing and implementing its own data collection platform. Such an 'institutional upgrade' is necessary in order to adequately meet the challenges of the digital economy and increase its effectiveness through the use of new techniques stemming from computational means. Below we present HCC's initiatives in this field, which include the development of computational tools, the establishment of a dedicated unit, and the dissemination of first-class research and advocacy efforts in this field.¹⁷

Computational tools for the detection of cartels, excessive and exclusionary pricing - developing the tools

The HCC commissioned a report to develop a screening method to detect anti-competitive practices from the analysis of market data and, in particular prices, taking advantage of new legislation enabling the authority to have mandated access to primary data regarding prices by the main supermarkets in the country, the distribution system for petrol stations, and the Athens central market for vegetables and fruits. This enables the authority to follow daily the level of prices for more than 2000 product codes across the country and to be able to use a time series since January 2020 and for some products a few years earlier.

The screen design took into account the following factors; broad applicability, low implementation cost, and accuracy. With respect to its methodology, the screen proposed was based on standard industrial economics principles and builds on top of existing screening methods to detect cartels and was tested using simulation techniques. A two-step method was followed comprising an initial screen to identify suspicious changes in pricing behavior, followed by a verification step to confirm whether the change is indeed consistent with anti-competitive conduct or whether other market events can explain it. The first step of the method consists of a preliminary analysis of unusual and suspicious price changes using a log diff-in-diff model. The second step of the method consists of a more in-depth industry-specific analysis, which involves estimating a pricing regression using supply- and demand-side variables as regressors and testing for structural breaks in the series. This way, it is only necessary to carry out a more complex analysis and to collect non-price data in case of suspected anti-competitive behavior during the previous stage. With respect to data requirements, the first step of the screen requires the collection of pricing data in a panel format. The second step of the method, apart from relying on pricing data, also requires the collection of supply- and demand-

¹⁷ For a detailed account of the HCC's engagement with computational tools as well as its broader ambition to explore ways the implementation of these new technologies may change the direction of competition law see its "inception" report entitled "Computational Competition Law and Economics - An Inception Report" <https://perma.cc/P9G6-6Z44>.

side data to use as controls. The proposed method has also successfully identified changes in competitive behavior in several simulation tests, performing better than alternative versions of the method.

→ HCC Economic Intelligence Platform

The HCC Intelligence Platform is an effort to integrate and keep updated multiple external data sources in a common database schema and provide visualization tools for data exploration and screening. The platform is being hosted within the premises of the HCC and is accessible for users with appropriate access credentials.¹⁸

The HCC Intelligence Platform allows the identification of the trends in category and product prices, for every firm or multiple firms and the monitoring of price changes day-by-day or week-by-week (time series options).

A more informative view of the price changes per category or product can be taken by using measures such as the percentiles, median, security threshold, and outliers. Each price is used as a unique data point in the analysis (not an average price of the week) (box plot options). The platform also allows for basket plot analysis. The platform provides a two-step screening tool whereby, as a first step, it conducts a preliminary analysis of unusual and suspicious price changes. As a second step, it tests for structural breaks in the price series.

A special mention of the HCC Intelligence Platform was made in the context of a research project on the use of new digital tools in the implementation of competition supported by the United Nations. The Competition Authorities of Brazil, Russia, and South Africa collaborated with the Hellenic Competition Commission in this program. This gave the Hellenic Competition Commission the opportunity to further develop this innovative tool in the future and to promote Greek know-how internationally.

→ Application

The above screening tool was implemented in a number of investigations during the first months of the Covid-19 pandemic. In particular, it was employed in the following investigations:

- health and hospital equipment procurement;
- markets for basic food commodities;
- healthcare materials.

¹⁸ Link: <https://perma.cc/L6LW-LWQF>.

Digital structure “upgrade”

Since October 2020, the HCC has established a forensic IT unit, which is headed by an economist and cooperates with a number of data scientists, who are acting as external experts for the authority. In addition, the HCC has appointed a Chief Economist and a Chief Technology Officer.

Next Steps: Establish a Public Ecosystem for Promoting Investments in Applied Complexity Science and Computational Competition Law and Economics

The HCC is in the process of promoting a network of international academic and competition authorities collaborations in order to establish a new public authority-academia platform for the study of complexity science and the development of corresponding computational tools (e.g., agent-based modeling, advanced simulations) in its economy mapping and enforcement activity, with the aim to develop concrete tools that could be of use by other competition authorities members, in the EU and beyond. The broader theoretical framework of this research program was described in the President of the HCC professor Ioannis Lianos’ article published in 2022.